Dirac equation 1

Consider the Schrodinger equation

$$i\hbar\frac{\partial}{\partial t}\psi(\mathbf{x},t) = H\psi(\mathbf{x},t)$$

The Hamiltonian H for the Dirac equation is

$$H = c\boldsymbol{\alpha}\cdot\mathbf{p} + \beta mc^2$$

where $\boldsymbol{\alpha} = (\alpha_x, \alpha_y, \alpha_z)$. The components of $\boldsymbol{\alpha}$ are the matrices

$$\alpha_x = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix} \quad \alpha_y = \begin{pmatrix} 0 & 0 & 0 & -i \\ 0 & 0 & i & 0 \\ 0 & -i & 0 & 0 \\ i & 0 & 0 & 0 \end{pmatrix} \quad \alpha_z = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{pmatrix}$$

and β is the matrix

$$\beta = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}$$

Substitute H into the Schrodinger equation.

$$i\hbar\frac{\partial}{\partial t}\psi(\mathbf{x},t) = (c\boldsymbol{\alpha}\cdot\mathbf{p} + \beta mc^2)\psi(\mathbf{x},t)$$

Rearrange as

$$\left(i\hbar\frac{\partial}{\partial t} - c\boldsymbol{\alpha}\cdot\mathbf{p}\right)\psi(\mathbf{x},t) = \beta mc^2\psi(\mathbf{x},t)$$

Substitute $\mathbf{p} = -i\hbar\nabla$ for the momentum operator.

$$i\hbar\left(\frac{\partial}{\partial t} + c\alpha_x\frac{\partial}{\partial x} + c\alpha_y\frac{\partial}{\partial y} + c\alpha_z\frac{\partial}{\partial z}\right)\psi(\mathbf{x},t) = \beta mc^2\psi(\mathbf{x},t)$$

Left-multiply by β.

$$i\hbar\left(\beta\frac{\partial}{\partial t} + c\beta\alpha_x\frac{\partial}{\partial x} + c\beta\alpha_y\frac{\partial}{\partial y} + c\beta\alpha_z\frac{\partial}{\partial z}\right)\psi(\mathbf{x},t) = mc^2\psi(\mathbf{x},t)$$

Define γ matrices

$$\gamma_t = \beta \quad \gamma_x = \beta\alpha_x \quad \gamma_y = \beta\alpha_y \quad \gamma_z = \beta\alpha_z$$

and write the Dirac equation

$$i\hbar\left(\gamma_t\frac{\partial}{\partial t} + c\gamma_x\frac{\partial}{\partial x} + c\gamma_y\frac{\partial}{\partial y} + c\gamma_z\frac{\partial}{\partial z}\right)\psi(\mathbf{x},t) = mc^2\psi(\mathbf{x},t)$$

In spacetime notation with $c = 1$

$$i\hbar\left(\gamma^0\frac{\partial}{\partial x^0}+\gamma^1\frac{\partial}{\partial x^1}+\gamma^2\frac{\partial}{\partial x^2}+\gamma^3\frac{\partial}{\partial x^3}\right)\psi(x^\mu)=m\psi(x^\mu)$$

Verify dimensions.

$$\hbar\frac{\partial}{\partial t}=[\mathrm{J\,s}]\,[\mathrm{s}^{-1}]=[\mathrm{J}]$$

$$\hbar c\frac{\partial}{\partial x}=[\mathrm{J\,s}]\,[\mathrm{m\,s}^{-1}]\,[\mathrm{m}^{-1}]=[\mathrm{J}]$$

$$mc^2=[\mathrm{J}]$$

Eigenmath script